# Grid Inverter ModBus RTU Protocol v1.00

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# 1. Basic information

From the communication point of view, Melacs Inverter is a SLAVE device type and allow to control it from any MASTER device working based on Modbus-RTU protocol stack.

Modbus-RTU USART driver implemented in Melacs Inverter device is flexible and support both popular physical serial bus interfaces:

- RS422 interface – in full duplex mode (due to echo suppression feature)
- RS485 interface – in half duplex mode

Modbus-RTU protocol implemented in Melacs Inverter is based on FreeMODBUS stack and allow to:

- Detect device on a bus (Report Slave ID function)
- Control the device, diagnose it and read all parameters from it (other supported functions)

Despite the fact that Modbus-RTU normally uses one of these USART configurations (always the same quantity of bits in each character):

- 1 start bit, 8 data bits, 1 parity bit (odd or even), 1 stop bit
- 1 start bit, 8 data bits, no parity bit, 2 stop bits

for properly cooperation with the broadest range of USB-to-USART bridges, this configuration presented below was selected at the project developing stage:

- 1 start bit, 8 data bits, no parity bit, 1 stop bit

and also the 115200 kbps speed was selected in current implementation.
These parameters can be easily changed in configuration files.

# 2. Function Codes List

## *2.1. Public Function Codes List (from Modbus specification)*

Below is presented table with a list of all Public Functions:

| Function Types | Function Data Sizes | | Function Names | Function Codes | | |
|---|---|---|---|---|---|---|
| | | | | code (dec) | subcode (dec) | code (hex) |
| Data Access | 1-Bit Access | Physical Discrete Inputs | Read Discrete Inputs | 02 | | 0x02 |
| | | Internal Bits or Physical Coils | Read Coils | 01 | | 0x01 |
| | | | Write Single Coil | 05 | | 0x05 |
| | | | Write Multiple Coils | 15 | | 0x0F |
| | 16-Bits Access | Physical Input Registers | Read Input Register | 04 | | 0x04 |
| | | Internal Registers or Physical Output Registers | Read Holding Registers | 03 | | 0x03 |
| | | | Write Single Register | 06 | | 0x06 |
| | | | Write Multiple Registers | 16 | | 0x10 |
| | | | Read/Write Multiple Registers | 23 | | 0x17 |
| | | | Mask Write Register | 22 | | 0x16 |
| | | | Read FIFO Queue | 24 | | 0x18 |
| | File Record Access | | Read File Record | 20 | | 0x14 |
| | | | Write File Record | 21 | | 0x15 |
| Diagnostics | | | Read Exception Status | 07 | | 0x07 |
| | | | Diagnostic | 08 | 00÷18,20 | 0x08 |
| | | | Get Com Event Counter | 11 | | 0x0B |
| | | | Get Com Event Log | 12 | | 0x0C |
| | | | Report Slave ID | 17 | | 0x11 |
| | | | Read Device Identification | 43 | | 0x2B |
| Other | | | Encapsulated Interface Transport | 43 | | 0x2B |

**Note:**
Based on the information from http://www.Modbus-IDA.org

## *2.2. Public Function Codes List (implemented)*

Below is presented table with a list of only implemented Public Functions:

| Function Types | Function Data Sizes | | Function Names | Function Codes & Names | | |
|---|---|---|---|---|---|---|
| | | | | code (dec) | code (hex) | user functions (name) |
| **Data Access** | **1-Bit Access** | Physical Discrete Inputs | Read Discrete Inputs | 02 | 0x02 | OCP |
| | | Internal Bits or Physical Coils | Read Coils | 01 | 0x01 | RLY<br>STATUS<br>RST |
| | | | Write Single Coil | 05 | 0x05 | RLY<br>SYS<br>REC |
| | | | Write Multiple Coils | 15 | 0x0F | |
| | **16-Bits Access** | Physical Input Registers | Read Input Register | 04 | 0x04 | ADC |
| | | Internal Registers or Physical Output Registers | Read Holding Registers | 03 | 0x03 | CALIB<br>HW_FW<br>MODE<br>DDS<br>ITRIP<br>DPLL<br>SM<br>REC_INFO<br>REC_DATA |
| | | | Write Single Register | 06 | 0x06 | MODE<br>DDS<br>ITRIP<br>DPLL |
| | | | Write Multiple Registers | 16 | 0x10 | |
| **Diagnostics** | | | Report Slave ID | 17 | 0x11 | |

# 3. Data format

## *3.1. Used Data Types*

| Name | Description | Size (bits / bytes) | Range |
|---|---|---|---|
| BOOL | 1-bit | 1 | 0 ÷ 1 |
| BYTE | 1-byte | 1 | 0 ÷ 255 |
| WORD | 2-bytes unsigned | 2 | 0 ÷ 65535 |
| INT | 2-bytes signed | 2 | -32768 ÷ 32767 |
| DWORD | 4-bytes unsigned | 4 | 0 ÷ 4294967296 |
| DINT | 4-bytes signed | 4 | -2147483648 ÷ 2147483647 |

# 4. Registers and Functions Description

## *4.1. OCP inputs states*

**Function 02 (0x02) – Read Discrete Inputs**

- **Read OCP Detector inputs states**

| (01000) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | OCP_OC_L1_INV | R | BOOL | L1_INV input state of OCP Detector |
| 00001 | OCP_OC_L2_INV | R | BOOL | L2_INV input state of OCP Detector |
| 00002 | OCP_OC_L3_INV | R | BOOL | L3_INV input state of OCP Detector |
| 00003 | OCP_OC_L1_GRID | R | BOOL | L1_GRID input state of OCP Detector |
| 00004 | OCP_OC_L2_GRID | R | BOOL | L2_GRID input state of OCP Detector |
| 00005 | OCP_OC_L3_GRID | R | BOOL | L3_GRID input state of OCP Detector |

## *4.2. RLY outputs states, RST sources & system STATUS*

**Function 01 (0x01) – Read Coils**

- **Read RLY outputs states**

| (02000) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | RELAY_INV_PHASE | R | BOOL | INV_PHASE relay output state |
| 00001 | RELAY_INV_NEUTRAL | R | BOOL | INV_NEUTRAL relay output state |
| 00002 | RELAY_GRID_PHASE | R | BOOL | GRID_PHASE relay output state |
| 00003 | RELAY_GRID_NEUTRAL | R | BOOL | GRID_NEUTRAL relay output state |

- **Read system STATUS (SYS, REC & SM)**

| (02100) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | SYS_RELOADED | R | BOOL | SYS_RELOADED flag state |
| 00001 | SYS_SHUTDOWN | R | BOOL | SYS_SHUTDOWN flag state |
| 00002 | | | | |
| 00003 | | | | |
| 00004 | | | | |
| 00005 | | | | |
| 00006 | | | | |
| 00007 | | | | |
| 00008 | REC_DATA_READY | R | BOOL | REC_DATA_READY flag state |
| 00009 | REC_DATA_REREAD | R | BOOL | REC_DATA_REREAD flag state |
| 00010 | | | | |
| 00011 | | | | |
| 00012 | | | | |
| 00013 | | | | |
| 00014 | | | | |
| 00015 | | | | |
| 00016 | SM_REC_STATE | R | BOOL | SM_REC_STATE flag state |

**Note:**
REC & SM flags are for Debug purpose only.

- **Read RST sources**

| (02200) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | RST_POR | R | BOOL | RST_POR flag state |
| 00001 | RST_BOR | R | BOOL | RST_BOR flag state |
| 00002 | RST_IDLE | R | BOOL | RST_IDLE flag state |
| 00003 | RST_SLEEP | R | BOOL | RST_SLEEP flag state |
| 00004 | RST_WDTO | R | BOOL | RST_WDTO flag state |
| 00005 | RST_SWDTEN | R | BOOL | RST_SWDTEN flag state |
| 00006 | RST_SWR | R | BOOL | RST_SWR flag state |
| 00007 | RST_EXTR | R | BOOL | RST_EXTR flag state |
| 00008 | RST_VREGS | R | BOOL | RST_VREGS flag state |
| 00009 | RST_CM | R | BOOL | RST_CM flag state |
| 00010 | | | | |
| 00011 | RST_VREGSF | R | BOOL | RST_VREGSF flag state |
| 00012 | | | | |
| 00013 | | | | |
| 00014 | RST_IOPUWR | R | BOOL | RST_IOPUWR flag state |
| 00015 | RST_TRAPR | R | BOOL | RST_TRAPR flag state |

## *4.3. RLY outputs control, SYS & REC control*

**Function 05 (0x05) – Write Single Coil**
**Function 15 (0x0F) – Write Multiple Coils**

- **Set RLY outputs state**

| (02000) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | RELAY_INV_PHASE | W | BOOL | INV_PHASE relay output state |
| 00001 | RELAY_INV_NEUTRAL | W | BOOL | INV_NEUTRAL relay output state |
| 00002 | RELAY_GRID_PHASE | W | BOOL | GRID_PHASE relay output state |
| 00003 | RELAY_GRID_NEUTRAL | W | BOOL | GRID_NEUTRAL relay output state |

- **Set SYS flags state**

| (02100) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | SYS_RELOADED | W | BOOL | SYS_RELOADED flag state |
| 00001 | SYS_SHUTDOWN | W | BOOL | SYS_SHUTDOWN flag state |
| 00002 | SYS_CALIB | W | BOOL | SYS_CALIB flag state |
| 00003 | SYS_RESET | W | BOOL | SYS_RESET flag state |

- **Set REC flags state**

| (02108) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | REC_DATA_REINIT | W | BOOL | REC_DATA_REINIT flag state |
| 00001 | REC_DATA_REREAD | W | BOOL | REC_DATA_REREAD flag state |

## *4.4. ADC Readings*

**Function 04 (0x04) – Read Input Register(s)**

- **ADC Readings**

| (03000) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | L1_U_INV_RAW | R | WORD | L1_U_INV value in ADC steps units |
| 00001 | L1_U_INV_MEAN | R | INT | L1_U_INV mean value in 1/32 V units |
| 00002 | L1_U_INV_PKPK | R | INT | L1_U_INV pk-pk value in 1/32 V units |
| 00003 | L1_U_INV_RMS | R | INT | L1_U_INV RMS value in 1/32 V units |
| 00004 | L1_I_INV_RAW | R | WORD | L1_I_INV value in ADC steps units |
| 00005 | L1_I_INV_MEAN | R | INT | L1_I_INV mean value in 1/32 A units |
| 00006 | L1_I_INV_PKPK | R | INT | L1_I_INV pk-pk value in 1/32 A units |
| 00007 | L1_I_INV_RMS | R | INT | L1_I_INV RMS value in 1/32 A units |
| 00008 | L2_U_INV_RAW | R | WORD | L2_U_INV value in ADC steps units |
| 00009 | L2_U_INV_MEAN | R | INT | L2_U_INV mean value in 1/32 V units |
| 00010 | L2_U_INV_PKPK | R | INT | L2_U_INV pk-pk value in 1/32 V units |
| 00011 | L2_U_INV_RMS | R | INT | L2_U_INV RMS value in 1/32 V units |
| 00012 | L2_I_INV_RAW | R | WORD | L2_I_INV value in ADC steps units |
| 00013 | L2_I_INV_MEAN | R | INT | L2_I_INV mean value in 1/32 A units |
| 00014 | L2_I_INV_PKPK | R | INT | L2_I_INV pk-pk value in 1/32 A units |
| 00015 | L2_I_INV_RMS | R | INT | L2_I_INV RMS value in 1/32 A units |
| 00016 | L3_U_INV_RAW | R | WORD | L3_U_INV value in ADC steps units |
| 00017 | L3_U_INV_MEAN | R | INT | L3_U_INV mean value in 1/32 V units |
| 00018 | L3_U_INV_PKPK | R | INT | L3_U_INV pk-pk value in 1/32 V units |
| 00019 | L3_U_INV_RMS | R | INT | L3_U_INV RMS value in 1/32 V units |
| 00020 | L3_I_INV_RAW | R | WORD | L3_I_INV value in ADC steps units |
| 00021 | L3_I_INV_MEAN | R | INT | L3_I_INV mean value in 1/32 A units |
| 00022 | L3_I_INV_PKPK | R | INT | L3_I_INV pk-pk value in 1/32 A units |
| 00023 | L3_I_INV_RMS | R | INT | L3_I_INV RMS value in 1/32 A units |
| 00024 | L1_U_MID_RAW | R | WORD | L1_U_MID value in ADC steps units |
| 00025 | L1_U_MID_MEAN | R | INT | L1_U_MID mean value in 1/32 V units |
| 00026 | L1_U_MID_PKPK | R | INT | L1_U_MID pk-pk value in 1/32 V units |
| 00027 | L1_U_MID_RMS | R | INT | L1_U_MID RMS value in 1/32 V units |
| 00028 | L2_U_MID_RAW | R | WORD | L2_U_MID value in ADC steps units |
| 00029 | L2_U_MID_MEAN | R | INT | L2_U_MID mean value in 1/32 V units |
| 00030 | L2_U_MID_PKPK | R | INT | L2_U_MID pk-pk value in 1/32 V units |
| 00031 | L2_U_MID_RMS | R | INT | L2_U_MID RMS value in 1/32 V units |
| 00032 | L3_U_MID_RAW | R | WORD | L3_U_MID value in ADC steps units |
| 00033 | L3_U_MID_MEAN | R | INT | L3_U_MID mean value in 1/32 V units |
| 00034 | L3_U_MID_PKPK | R | INT | L3_U_MID pk-pk value in 1/32 V units |
| 00035 | L3_U_MID_RMS | R | INT | L3_U_MID RMS value in 1/32 V units |

| | | | | |
|---|---|---|---|---|
| 00036 | L1_U_GRID_RAW | R | WORD | L1_U_GRID value in ADC steps units |
| 00037 | L1_U_GRID_MEAN | R | INT | L1_U_GRID mean value in 1/32 V units |
| 00038 | L1_U_GRID_PKPK | R | INT | L1_U_GRID pk-pk value in 1/32 V units |
| 00039 | L1_U_GRID_RMS | R | INT | L1_U_GRID RMS value in 1/32 V units |
| 00040 | L1_I_GRID_RAW | R | WORD | L1_I_GRID value in ADC steps units |
| 00041 | L1_I_GRID_MEAN | R | INT | L1_I_GRID mean value in 1/32 A units |
| 00042 | L1_I_GRID_PKPK | R | INT | L1_I_GRID pk-pk value in 1/32 A units |
| 00043 | L1_I_GRID_RMS | R | INT | L1_I_GRID RMS value in 1/32 A units |
| 00044 | L2_U_GRID_RAW | R | WORD | L2_U_GRID value in ADC steps units |
| 00045 | L2_U_GRID_MEAN | R | INT | L2_U_GRID mean value in 1/32 V units |
| 00046 | L2_U_GRID_PKPK | R | INT | L2_U_GRID pk-pk value in 1/32 V units |
| 00047 | L2_U_GRID_RMS | R | INT | L2_U_GRID RMS value in 1/32 V units |
| 00048 | L2_I_GRID_RAW | R | WORD | L2_I_GRID value in ADC steps units |
| 00049 | L2_I_GRID_MEAN | R | INT | L2_I_GRID mean value in 1/32 A units |
| 00050 | L2_I_GRID_PKPK | R | INT | L2_I_GRID pk-pk value in 1/32 A units |
| 00051 | L2_I_GRID_RMS | R | INT | L2_I_GRID RMS value in 1/32 A units |
| 00052 | L3_U_GRID_RAW | R | WORD | L3_U_GRID value in ADC steps units |
| 00053 | L3_U_GRID_MEAN | R | INT | L3_U_GRID mean value in 1/32 V units |
| 00054 | L3_U_GRID_PKPK | R | INT | L3_U_GRID pk-pk value in 1/32 V units |
| 00055 | L3_U_GRID_RMS | R | INT | L3_U_GRID RMS value in 1/32 V units |
| 00056 | L3_I_GRID_RAW | R | WORD | L3_I_GRID value in ADC steps units |
| 00057 | L3_I_GRID_MEAN | R | INT | L3_I_GRID mean value in 1/32 A units |
| 00058 | L3_I_GRID_PKPK | R | INT | L3_I_GRID pk-pk value in 1/32 A units |
| 00059 | L3_I_GRID_RMS | R | INT | L3_I_GRID RMS value in 1/32 A units |
| 00060 | P_U_DC_RAW | R | WORD | P_U_DC value in ADC steps units |
| 00061 | P_U_DC_MEAN | R | WORD | P_U_DC mean value in 1/32 V units |
| 00062 | P_U_DC_PKPK | R | WORD | P_U_DC pk-pk value in 1/32 V units |
| 00063 | P_U_DC_RMS | R | WORD | P_U_DC RMS value in 1/32 V units |
| 00064 | N_U_DC_RAW | R | WORD | N_U_DC value in ADC steps units |
| 00065 | N_U_DC_MEAN | R | WORD | N_U_DC mean value in 1/32 V units |
| 00066 | N_U_DC_PKPK | R | WORD | N_U_DC pk-pk value in 1/32 V units |
| 00067 | N_U_DC_RMS | R | WORD | N_U_DC RMS value in 1/32 V units |
| 00068 | T1_RAW | R | WORD | T1_RAW value in ADC steps units |
| 00069 | T1_CELSIUS | R | INT | T1_CELSIUS value in 1/10 °C units |
| 00070 | T1_RESISTANCE | R | DWORD | T1_RESISTANCE value in 1/10 Ω units |
| 00072 | T2_RAW | R | WORD | T2_RAW value in ADC steps units |
| 00073 | T2_CELSIUS | R | INT | T2_CELSIUS value in 1/10 °C units |
| 00074 | T2_RESISTANCE | R | DWORD | T2_RESISTANCE value in 1/10 Ω units |
| 00076 | T3_RAW | R | WORD | T3_RAW value in ADC steps units |
| 00077 | T3_CELSIUS | R | INT | T3_CELSIUS value in 1/10 °C units |
| 00078 | T3_RESISTANCE | R | DWORD | T3_RESISTANCE value in 1/10 Ω units |

## *4.5. CALIB & HW_FW Readings*

**Function 04 (0x04) – Read Input Register(s)**

- **CALIB Readings**

| (04000) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | L1_I_INV_RAW_ERR | R | INT | L1_I_INV_ERR value in ADC steps units |
| 00001 | L1_I_INV_UNIT_ERR | R | INT | L1_I_INV_ERR value in 1/32 A units |
| 00002 | L2_I_INV_RAW_ERR | R | INT | L2_I_INV_ERR value in ADC steps units |
| 00003 | L2_I_INV_UNIT_ERR | R | INT | L2_I_INV_ERR value in 1/32 A units |
| 00004 | L3_I_INV_RAW_ERR | R | INT | L3_I_INV_ERR value in ADC steps units |
| 00005 | L3_I_INV_UNIT_ERR | R | INT | L3_I_INV_ERR value in 1/32 A units |
| 00006 | L1_I_GRID_RAW_ERR | R | INT | L1_I_GRID_ERR value in ADC steps units |
| 00007 | L1_I_GRID_UNIT_ERR | R | INT | L1_I_GRID_ERR value in 1/32 A units |
| 00008 | L2_I_GRID_RAW_ERR | R | INT | L2_I_GRID_ERR value in ADC steps units |
| 00009 | L2_I_GRID_UNIT_ERR | R | INT | L2_I_GRID_ERR value in 1/32 A units |
| 00010 | L3_I_GRID_RAW_ERR | R | INT | L3_I_GRID_ERR value in ADC steps units |
| 00011 | L3_I_GRID_UNIT_ERR | R | INT | L3_I_GRID_ERR value in 1/32 A units |
| 00012 | L1_U_INV_RAW_ERR | R | INT | L1_U_INV_ERR value in ADC steps units |
| 00013 | L1_U_INV_UNIT_ERR | R | INT | L1_U_INV_ERR value in 1/32 V units |
| 00014 | L2_U_INV_RAW_ERR | R | INT | L2_U_INV_ERR value in ADC steps units |
| 00015 | L2_U_INV_UNIT_ERR | R | INT | L2_U_INV_ERR value in 1/32 V units |
| 00016 | L3_U_INV_RAW_ERR | R | INT | L3_U_INV_ERR value in ADC steps units |
| 00017 | L3_U_INV_UNIT_ERR | R | INT | L3_U_INV_ERR value in 1/32 V units |
| 00018 | L1_U_MID_RAW_ERR | R | INT | L1_U_MID_ERR value in ADC steps units |
| 00019 | L1_U_MID_UNIT_ERR | R | INT | L1_U_MID_ERR value in 1/32 V units |
| 00020 | L2_U_MID_RAW_ERR | R | INT | L2_U_MID_ERR value in ADC steps units |
| 00021 | L2_U_MID_UNIT_ERR | R | INT | L2_U_MID_ERR value in 1/32 V units |
| 00022 | L3_U_MID_RAW_ERR | R | INT | L3_U_MID_ERR value in ADC steps units |
| 00023 | L3_U_MID_UNIT_ERR | R | INT | L3_U_MID_ERR value in 1/32 V units |
| 00024 | L1_U_GRID_RAW_ERR | R | INT | L1_U_GRID_ERR value in ADC steps units |
| 00025 | L1_U_GRID_UNIT_ERR | R | INT | L1_U_GRID_ERR value in 1/32 V units |
| 00026 | L2_U_GRID_RAW_ERR | R | INT | L2_U_GRID_ERR value in ADC steps units |
| 00027 | L2_U_GRID_UNIT_ERR | R | INT | L2_U_GRID_ERR value in 1/32 V units |
| 00028 | L3_U_GRID_RAW_ERR | R | INT | L3_U_GRID_ERR value in ADC steps units |
| 00029 | L3_U_GRID_UNIT_ERR | R | INT | L3_U_GRID_ERR value in 1/32 V units |

- **HW_FW Readings**

| (04100) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | HW_INFO | R | WORD | major & minor HW version number |
| 00001 | FW_VERSION | R | WORD | major & minor FW version number |
| 00002 | FW_DATE | R | WORD | FW date - year |
| 00003 | FW_MONTH_DAY | R | WORD | FW date – month & day |

## *4.6. MODE, DDS, ITRIP & DPLL Readings*

**Function 04 (0x04) – Read Input Register(s)**

- **MODE Readings**

| (04200) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | POWERUP_COUNTER | R | DWORD | Timer ticks since powerup in 1s units |
| 00002 | SM_SYSTEM_MODE | R | BYTE | System mode {look at Table 01} |

- **DDS Readings**

| (04300) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | WAVES_FREQ | R | WORD | Frequency in 1/10 Hz units |
| 00001 | SHIFT_PHASE_CH2 | R | INT | Phase shift for CH2 in 1/10 ° DEG |
| 00002 | SHIFT_PHASE_CH3 | R | INT | Phase shift for CH3 in 1/10 ° DEG |
| 00003 | GAIN_CH1 | R | WORD | Amplitude for CH1 in range 0÷65535 |
| 00004 | GAIN_CH2 | R | WORD | Amplitude for CH2 in range 0÷65535 |
| 00005 | GAIN_CH3 | R | WORD | Amplitude for CH3 in range 0÷65535 |

- **ITRIP Readings**

| (04400) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | ITRIP_INV_UPPER | R | WORD | ITRIP_INV_UPPER in range 0÷65535 |
| 00001 | ITRIP_GRID_LOWER | R | WORD | ITRIP_GRID_LOWER in range 0÷65535 |

- **DPLL Readings**

| (04500) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | ID_SET | R | INT | ID_SET in range -32768÷32767 |
| 00001 | IQ_SET | R | INT | IQ_SET in range -32768÷32767 |
| 00002 | VD_SET | R | INT | VD_SET in range -32768÷32767 |
| 00003 | VQ_SET | R | INT | VQ_SET in range -32768÷32767 |
| 00004 | OMEGA_L | R | INT | OMEGA_L in range -32768÷32767 |

## *4.7. SM, REC_INFO & REC_DATA Readings (for Debug purpose only)*

**Function 04 (0x04) – Read Input Register(s)**

- **SM Readings**

| (09000) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | SM_REC_STATE | R | BYTE | |
| 00001 | SM_STATE_ERROR | R | WORD | |
| 00002 | SM_TIMESTAMP_MS | R | WORD | |
| 00002 | SM_TIMESTAMP_S | R | DWORD | |

- **REC_INFO Readings**

| (09100) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | DATA_REC_LENGTH | R | DWORD | |

- **REC_INFO Readings**

| (09200) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | DATA_REC_ADDR | R | DWORD | |
| 00002 | DATA_REC_DATA | R | WORD | |
| … | DATA_REC_DATA | R | WORD | |
| 00125 | DATA_REC_DATA | R | WORD | |

## *4.8. MODE, DDS, ITRIP & DPLL Setup*

**Function 04 (0x04) – Read Input Register(s)**

- **MODE Setup**

| (04202) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00002 | SM_SYSTEM_MODE | W | BYTE | System mode {look at Table 01} |

- **DDS Readings**

| (04300) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | WAVES_FREQ | W | WORD | Frequency in 1/10 Hz units |
| 00001 | SHIFT_PHASE_CH2 | W | INT | Phase shift for CH2 in 1/10 ° DEG |
| 00002 | SHIFT_PHASE_CH3 | W | INT | Phase shift for CH3 in 1/10 ° DEG |
| 00003 | GAIN_CH1 | W | WORD | Amplitude for CH1 in range 0÷65535 |
| 00004 | GAIN_CH2 | W | WORD | Amplitude for CH2 in range 0÷65535 |
| 00005 | GAIN_CH3 | W | WORD | Amplitude for CH3 in range 0÷65535 |

- **ITRIP Readings**

| (04400) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | ITRIP_INV_UPPER | W | WORD | ITRIP_INV_UPPER in range 0÷65535 |
| 00001 | ITRIP_GRID_LOWER | W | WORD | ITRIP_GRID_LOWER in range 0÷65535 |

- **DPLL Readings**

| (04500) +Offset | Name | I/O | Type | Description |
|---|---|---|---|---|
| 00000 | ID_SET | W | INT | ID_SET in range -32768÷32767 |
| 00001 | IQ_SET | W | INT | IQ_SET in range -32768÷32767 |
| 00002 | VD_SET | W | INT | VD_SET in range -32768÷32767 |
| 00003 | VQ_SET | W | INT | VQ_SET in range -32768÷32767 |
| 00004 | OMEGA_L | W | INT | OMEGA_L in range -32768÷32767 |

## *4.9. Parameter Tables*

**Table 01. System Mode**

| Value | Description |
|---|---|
| 0 | Standalone 1-Phase |
| 1 | Standalone 3-Phases |
| 2 | Grid Connected 1-Phase - no BAT |
| 3 | Grid Connected 3-Phases - no BAT |
| 4 | Test |
| 5 | Grid Connected 1-Phase |
| 6 | Grid Connected 3-Phases |
| 7 | Additional Test |

**Table 02. SM State**

| Value | Description |
|---|---|
| 0 | 0 - PRE_START |
| 1 | 1 - START |
| 2 | 2 - IDLE |
| 3 | 3 - ERROR |
| 4 | 4 - GRID_RLY_CLOSE |
| 5 | 5 - DPLL_INIT |
| 6 | 6 - DPLL_LOCKED |
| 7 | 7 - INV_RLY_CLOSE |
| 8 | 8 - DELAY |
| 9 | 9 - TIMEOUT |
| 10 | 10 - WAIT_FOR_REINIT |
| 11 | 11 - ISLAND_MODE |
| 12 | 12 - ANTY_ISLANDING |
| 13 | 13 - C_RAMP_BEGIN |
| 14 | 14 - C_RAMP_END |

**Table 03. SM Error**

| Value | Description |
|---|---|
| 0 | 0 - NONE |
| 1 | 1 - CRITICAL |
| 2 | 2 - INITIALIZATION |
| 3 | 3 - NO_DPLL_LOCK |
| 4 | 4 - NO_GRID |
| 5 | 5 - OVERCURRENT |
| 6 | 6 - INV_VOLTAGE |
| 7 | 7 - MID_VOLTAGE |
| 8 | 8 - GRID_VOLTAGE |
| 9 | 9 - TIMEOUT |